

Customer Shopping Behavior Analysis

1. Project Overview:

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary:

- **Rows:** 3,900
- **Columns:** 18
- **Key Features:**
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount (USD), Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method)
- **Missing Data:** 37 values in Review Rating column

3. Exploratory Data Analysis Using Python:

We begin this project with data preparation and cleaning using Python.

- **Data Loading:** Importing the dataset using the Pandas library and displaying the first 5 rows (df.head()) to inspect the initial data structure.

```
[1]: import pandas as pd
[2]: df = pd.read_csv('customer_shopping_behavior.csv')
[3]: df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Ve
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	(
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	C
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	Pe
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Pe

- **Initial Explorations(EDA):** The next step is exploring the data using `df.info()` to check the structure and data types, and `df.describe().round(2)` to view statistical summaries formatted to two decimal places for easier analysis.

```
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                   3900 non-null   int64
2   Gender                               3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                  3900 non-null   object
8   Color                                 3900 non-null   object
9   Season                                3900 non-null   object
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                        3900 non-null   object
13  Discount Applied                     3900 non-null   object
14  Promo Code Used                      3900 non-null   object
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                       3900 non-null   object
17  Frequency of Purchases               3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
df.describe().round(2)
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.00	3900.00	3900.00	3863.00	3900.00
mean	1950.50	44.07	59.76	3.75	25.35
std	1125.98	15.21	23.69	0.72	14.45
min	1.00	18.00	20.00	2.50	1.00
25%	975.75	31.00	39.00	3.10	13.00
50%	1950.50	44.00	60.00	3.80	25.00
75%	2925.25	57.00	81.00	4.40	38.00
max	3900.00	70.00	100.00	5.00	50.00

- **Missing Values Check:** The next step is identifying missing data using `df.isnull().sum()`. The output reveals that there are 37 missing values specifically in the Review Rating column.

```
: df.isnull().sum()

: Customer ID      0
  Age             0
  Gender           0
  Item Purchased   0
  Category         0
  Purchase Amount (USD)  0
  Location         0
  Size            0
  Color           0
  Season          0
  Review Rating    37
  Subscription Status  0
  Shipping Type    0
  Discount Applied  0
  Promo Code Used  0
  Previous Purchases  0
  Payment Method   0
  Frequency of Purchases  0
dtype: int64
```

- **Missing Data Handling:** After identifying the missing data, the next step is to handle the null values in the Review Rating column. To prevent bias, we impute these gaps using the median rating of their respective product Category. A final check using `df.isnull().sum()` confirms that the dataset is now completely clean.

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))

df.isnull().sum()

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season           0
Review Rating    0
Subscription Status  0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

- **Column Standardization:** Renaming all columns to lowercase and replacing spaces with underscores to achieve a standard *snake_case* format. This ensures consistency, improves readability, and prevents syntax errors during SQL integration.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
```

- **Feature Engineering:** This stage focuses on creating new variables to enhance the analysis.
 - **Created age_group:** column to segment customer ages into distinct demographic categories (e.g., Young Adult, Adult, Middle-aged, and Senior). This was achieved by binning the ages into quartiles using `pd.qcut()`, ensuring an even distribution across groups. This step is essential for identifying specific purchasing patterns across different age demographics.

```
# Create a column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

- **Created purchase_frequency_days:** Converted categorical text values from the purchase frequency column (e.g., 'Weekly', 'Monthly') into their numerical day equivalents using a dictionary mapping technique. Transforming categorical strings into numerical data is an essential step that enables mathematical calculations, aggregations, and more advanced quantitative analysis of customer behavior.

```
# Create column purchase_frequency_days

frequency_mapping = {
    'Fortnightly' : 14,
    'Weekly' : 7,
    'Monthly' : 30,
    'Quarterly' : 90,
    'Bi-Weekly' : 14,
    'Annually' : 345,
    'Every 3 Months' : 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

- **Handling Redundant Features:**

Verified that the discount_applied and promo_code_used columns are entirely identical. Dropped the promo_code_used column to prevent data duplication, and confirmed its successful removal using the df.columns attribute.

```
] : (df['discount_applied'] == df['promo_code_used']).all()
] : np.True_
] : df = df.drop('promo_code_used', axis=1)
] : df.columns

] : Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'previous_purchases', 'payment_method',
        'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
        dtype='object')
```

- **Database Integration:**

In this phase, we establish a connection between the Python script and a PostgreSQL database using the SQLAlchemy library. Once the connection engine is set up, the fully cleaned dataset is loaded into a PostgreSQL table using the `df.to_sql()` function. This step bridges the data preparation phase with the analytical phase, readying the data for in-depth exploration using SQL queries.

```
from sqlalchemy import create_engine

# Step 1: Connect to PostgreSQL
username = "postgres"
password = "postgres"
host = "localhost"
port = "5432"
database = "customer_behavior"

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

# Step 2: Load DataFrame into PostgreSQL
table_name = "customer"
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data Successfully loaded into table '{table_name}' in database '{database}'. ")
Data Successfully loaded into table 'customer' in database 'customer_behavior'.
```

4. Data Analysis Using SQL

After successfully cleaning and preparing the dataset with Python, the next phase involves conducting in-depth data analysis using SQL. This step aims to extract actionable insights and answer strategic business questions based on customer transaction patterns.

1. Revenue by Gender:

This analysis compares the total revenue generated by male and female customers. The query results reveal that male customers contributed a significantly higher total revenue (157,890) compared to female customers (75,191).

	gender text	revenue_from_customers numeric
1	Female	75191
2	Male	157890

2. High-Value Discount Users:

This analysis identifies customers who applied a discount but still recorded a purchase amount strictly above the overall average. Uncovering this specific segment helps the business optimize future marketing campaigns by targeting high-yield customers rather than just bargain hunters.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
Total rows: 839 of 839		Query complete C

3. Top 5 Products by Average Rating:

This analysis identifies the top five products with the highest average customer ratings. Highlighting these top-performing items provides valuable insights into customer satisfaction and helps the business prioritize inventory and promotional efforts for highly favored products.

	item_purchased text	average_product_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison:

This analysis compares the average purchase amount across different shipping methods, specifically focusing on express versus standard delivery. Understanding this relationship helps the business determine if customers who opt for faster delivery tend to have higher cart values, providing valuable insights for logistics and pricing strategies.

	shipping_type text	average_purchase_amount numeric
1	Standard	58.46
2	Express	60.48

5. Subscriber vs. Non-Subscriber Analysis:

This step compares the customer volume, average spend, and total revenue between subscribed and non-subscribed customers. The query reveals an interesting insight: non-subscribers not only drive the majority of the total revenue, but they also have a marginally higher average spend (58.87) compared to subscribers (59.49). This critical finding suggests that the business may need to re-evaluate its current subscription program to ensure it effectively incentivizes higher spending.

	subscription_status text	total_customers bigint	average_spend numeric	total_revenue numeric
1	No	2847	59.87	170436
2	Yes	1053	59.49	62645

6. Discount-Dependent Products:

Identified 5 products with the highest percentage of discounted purchases. To evaluate a product's reliance on price reductions to drive sales. This insight is used to reassess base pricing strategies, protect profit margins from excessive discounting, and monitor the effectiveness of inventory clearance efforts.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer Segmentation:

This analysis categorizes customers into three distinct segments (New, Returning, and Loyal) based on their historical transaction frequency. The query results reveal a massive base of Loyal customers (3,116), indicating excellent user retention. However, the significantly low count of New customers (83) highlights a critical gap in user acquisition, signaling an urgent need to revamp marketing strategies to attract fresh users.

	customer_segment text	number_of_customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 Best-Selling Products per Category:

This analysis identifies the top three products with the highest order volumes within each category. The results highlight category leaders, such as Jewelry dominating Accessories, and Blouses and Pants leading the Clothing segment. Pinpointing these flagship items is crucial for the merchandising and supply chain teams to guarantee inventory availability and optimize category-specific marketing efforts.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Purchases and Subscription Correlation:

This analysis investigates whether frequent shoppers (those with more than 5 transactions) are more likely to opt into the subscription program. The results reveal a critical insight: the vast majority of these highly engaged repeat buyers (2,518 customers) remain non-subscribers. This represents a massive missed opportunity for the business, strongly indicating that the current subscription benefits fail to incentivize even the most active and loyal customers.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group:

This analysis aggregates total revenue to map purchasing power across different customer age segments. The query results reveal that the *Young Adult* demographic is the primary revenue driver (62,143), closely followed by the *Middle-aged* group (59,197). This distribution insight is highly actionable for the Marketing team, allowing them to optimize ad spend and tailor promotional campaigns toward the most profitable demographics.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

The final step of this project is summarizing all the data insights into an interactive dashboard using Power BI. The goal is to translate complex numbers into easy-to-understand visual charts. This dashboard highlights essential business metrics—such as total customers, average spending, and revenue by category. Equipped with filtering features, it allows the business team to easily explore the data and make quick, informed decisions.



6. Business Recommendations

- **Improve Subscription Benefits:** Our data shows that subscribers and regular customers spend about the same amount (~\$59). We need to offer better perks to encourage our frequent shoppers to join the subscription program and increase their spending.
- **Focus on Gaining New Customers:** We are doing a great job at keeping our current customers (with over 3,000 loyal ones), but we are getting very few new buyers. We should shift our marketing focus to actively attract fresh customers.
- **Review Discount Strategy:** Avoid offering massive 50% discounts too often on items like Hats and Sneakers. We can test offering smaller discounts to see if sales remain strong without losing too much profit.

- **Highlight Top Products:** Make our highest-rated and best-selling items the main stars of our marketing campaigns. Promoting these customer favorites will naturally drive sales without us having to constantly rely on low prices.
- **Target the Right Audience:** The "Young Adult" group brings in the most revenue. We should direct the majority of our advertising budget toward this specific age group to ensure our marketing efforts are highly effective and profitable.